

“Indian market and industries will grow to become huge and energetic. Indian chemical companies and R&D centres can play as one of the world leaders”



While talking about breakthrough technologies possible with the collaboration of electronics and chemicals, Dr Osamu Kumagai, SVP, corporate executive, president, advanced materials laboratories, president, core device development group, Sony Corporation, speaks to Uma Bansal, executive editor at EFY, about the future of advanced chemistry in India

■ **DR OSAMU KUMAGAI**
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What are the major breakthroughs in electronics which are made possible by chemicals recently?

Some commercially available technologies include:

1. LCD (liquid crystal materials are organic materials serving as electric light shutters)
 2. Organic LED display (organic materials serve as a light emitting materials)
 3. Electronic Paper (electrophoretic display)
- Technologies in developmental phase include flexible displays made by organic thin film transistors in which organic materials serve as semiconductors, and also dye-sensitised solar cells where organic dyes serve as photoelectric conversion elements.

What further potential do they have to change the way electronic products are designed and used?

Organic materials can be designed to be soluble as an ink through chemical synthesis. With an ink we can utilise the printing technology. This costs much less than the con-

ventional silicon process, and has a very high throughput. It does not need high temperature, so we can use plastic films as substrates for devices. All this combined we can produce organic or plastic-based flexible products; for example, flexible, thin, lightweight electronic paper just like real paper in the future.

Coming to LED displays and flexible TVs, how would these chemicals impact the product innovation by Sony?

The research and development (R&D) of organic and molecular materials is quite important in the field of organic electronics. 'Organic molecules' mean the compounds with carbon frames. We can have various properties by changing molecules through chemical reactions. For example, if we manage to add the solubility, such a chemical can be made to an ink. We can coat or print with this ink to make a device.

Sony develops new materials such as organic semiconductors and dyes through organic and molecular synthesis.